

CurryTracker

System Architecture, Multi-Tenant Design & Complete Creation Guide

Live App: <https://expense-tracker-77br.onrender.com> • Date: July 2026

1. Executive Summary

CurryTracker is a production-grade, multi-tenant web application designed for managing shared room expenses, daily curry/meal splits, UPI payment settlements, and roommate communication. Built with a lightweight Node.js/Express backend and an interactive Vanilla JavaScript Single-Page Application (SPA) frontend, CurryTracker provides zero-latency expense tracking, dynamic dynamic per-meal cost calculation, 2-second mobile push notifications, dynamic custom UPI QR code generation, and 1-tap Telegram group report broadcasting.

Feature	Implementation & Highlights
Multi-Tenant Security	3-tier RBAC: Main Super Admin, Secondary Group Admins, and Standard Members.
Dynamic Split Engine	Dynamic per-meal cost division based on participating member array (splitBetween).
Dual Storage Engine	Embedded SQLite3 synchronized with persistent JSON data files with auto-healing fallback.
Mobile Web Push	2000ms polling push engine delivering status bar alerts with sound and vibration patterns.
UPI & Telegram	Dynamic custom amount UPI QR generator (upiqr.in) & 1-tap Telegram markdown share.

2. System Architecture & Technology Stack

Layer	Technology / Framework	Core Purpose
Frontend Core	HTML5, Vanilla JS (ES6+ SPA)	High performance 60fps UI, modular view routing without heavy framework bundles.

UI & Aesthetics	Glassmorphism, CSS Tokens, Bootstrap 5	Vibrant dark/light hybrid design system, responsive modals, animated stat cards.
Analytics & Charts	Chart.js v4	Interactive doughnut pie charts, monthly line trends, and weekly bar charts.
Backend Framework	Node.js, Express.js	RESTful API architecture, route handling, express-session cookie management.
Security & Auth	Bcrypt.js, Express-Session, RBAC	Salted password hashing (factor 10), session isolation, role enforcement middleware.
Database Layer	SQLite3 + Synchronized JSON Data Files	Dual-storage fallback engine ensuring zero data loss across cloud deployments.
Real-Time Engine	Mobile Push API + 2000ms Polling	Instant status bar push notifications with sound, vibration, and custom icons.
Integrations	upiqr.in API, Telegram Web API	Dynamic custom amount UPI QR generation & 1-tap Telegram group report sharing.

3. Multi-Tenant Role-Based Access Control (RBAC)

CurryTracker enforces a strict 3-tier security and visibility hierarchy across all API endpoints and frontend views:

Role Tier	Primary Account Example	Scope & System Permissions
■ Main Super Admin super_admin	Kandukuri Jagan (User ID: 192472374)	• Full platform-wide master dashboard & global expense analytics. • Exclusive ability to create Secondary Group Admins. • Exclusive access to Credentials Directory with password unhide toggles. • Exclusive authority to remove/delete (■■■) member accounts.
■■ Secondary Group Admin admin	anudeep (User ID: 192472066)	• Isolated dashboard and expense manager dedicated ONLY to their room group. • Can add regular Member accounts to their group. • Record expenses & cash settlements for their room. • Hidden from credentials directory & super admin settings.

■ Standard Member member	Sagar, Prathap, Bharath, Charan, Ganesh, a	• Isolated view restricted ONLY to expenses where they paid or participated in the split. • Live balance card ('■ Owes', '■ Receives', '■ Settled'). • Submit payment settlement records.
------------------------------------	---	---

4. Core Mathematical Algorithms & System Logic

4.1 Dynamic Per-Meal Split Formula

Unlike basic expense apps that assume fixed 4-way splits, CurryTracker dynamically computes per-meal shares based on the specific participating members selected (splitBetween):

```
Share per Person (Meal i) = Amount(i) / Count(splitBetween_i)

Total Paid (Member m) = Σ Amount(i) where paidBy = m
Total Share (Member m) = Σ (Amount(i) / Count(splitBetween_i)) where m ∈ splitBetween_i

Net Balance (Member m) = (Total Paid + Settlements Out) - (Total Share + Settlements In)
```

4.2 Auto-Healing Persistent Dual-Storage Engine

To guarantee zero data loss across cloud deployments (such as Render ephemeral container restarts), CurryTracker operates a dual-storage engine:

- **Primary Engine:** Embedded SQLite3 database executing SQL transactions for high-speed queries.
- **Backup Persistence Engine:** State synchronized to formatted JSON files (data/users.json, data/expenses.json, data/settlements.json).
- **Auto-Healing Initialization:** On server startup, initDatabase() verifies seed records and self-heals any missing user accounts or sample expenses.

5. Codebase Taxonomy & Application Structure

```
expense-tracker/
■■■ server.js • App entry point, Express server initialization, session middleware, route mounting.
■■■ database.js • SQLite3 database connection, JSON read/write handlers, schema setup & seed data.
■■■ middleware/auth.js • RBAC protection guards (requireAuth, requireAdmin, requireSuperAdmin).
■■■ routes/
■ ■■■ auth.js • User sign-in, session verification, logout endpoints.
■ ■■■ members.js • Member management, Group Admin creation, credentials directory, user removal.
■ ■■■ expenses.js • Expense CRUD, dynamic split parsing, group filtering.
■ ■■■ balance.js • Group balance calculation engine, settlement recording, cash clearing.
■ ■■■ dashboard.js • Aggregated statistics & Chart.js data formatting.
■ ■■■ messages.js • Roommate chatroom & Admin broadcast banner backend.
■ ■■■ reports.js • Monthly analytics & CSV/Excel/PDF export generator.
■■■ public/
```

■■■ **dashboard.html** • SPA single HTML shell containing glassmorphic view containers & Bootstrap modals.

■■■ **css/style.css** • CSS Design System, tokens, glassmorphism utilities, dark/light theme variables.

■■■ **js/**

■■■ **app.js** • SPA Router, session state manager, view navigation guards.

■■■ **dashboard.js** • Dashboard view renderer, counter animations, Chart.js instances.

■■■ **members.js** • Members cards rendering, login directory, Group Admin modal.

■■■ **expenses.js** • Expense history table, Add/Edit modal handlers.

■■■ **payments.js** • Payment balance sheet, settlements table, custom amount UPI QR generator.

■■■ **messages.js** • Roommate chat & Broadcast banner UI controllers.

■■■ **notifications.js** • 2-second ultra-fast mobile push notification manager.

6. Key Features & External Integrations

6.1 Dynamic Custom Amount UPI QR Generator

Integrated dynamic UPI QR code generator powered by the upiqr.in API. Users can input any custom settlement amount or click their exact outstanding balance ('■ Owes'). Generates an instant scannable UPI QR code targeting verified VPA 8367047947@ybl with direct deep links to PhonePe, Google Pay, and Paytm.

6.2 2-Second Mobile Push Notification Engine

Features a high-frequency (2000ms interval) browser notification synchronizer. Native push notifications pop up directly in the Android/iOS status bar with custom sound, vibration patterns ([500, 200, 500, 200, 500]), and app crown icons whenever expenses, payment nudges, or messages are posted.

6.3 1-Tap Telegram Group Report Share

Includes 1-tap **Share on Telegram** buttons across Dashboard, Reports, and Payments views. Automatically compiles an itemized Markdown report (total curry bills, per-person share, member-by-member breakdown, UPI ID) and opens Telegram Web or Desktop app via <https://t.me/share/url> to broadcast directly to room groups.

7. Deployment & Operational Credentials

Parameter	Details / Value
Live Hosted URL	https://expense-tracker-77br.onrender.com
GitHub Repository	https://github.com/endukuniku35-ctrl/expense-tracker.git
Main Super Admin Account	User ID: 192472374 • Password: kandukurijagan@14062020 (Kandukuri Jagan)
Secondary Group Admin Account	User ID: anudeep • Password: anudeep (Group Admin Flat 301)

Standard Member Accounts	User IDs: 192472343 (Sagar), 192411184 (Prathap), 192411185 (Bharath), 192412348 (Charan), Ganesh (Ganesh), a (Member a)
Default Payment VPA	UPI ID: 8367047947@ybl

Documentation Verification & Sign-off

This architectural guide represents the complete, verified state of the CurryTracker multi-tenant expense management platform. All features, security policies, role isolations, and integrations are live and fully operational.